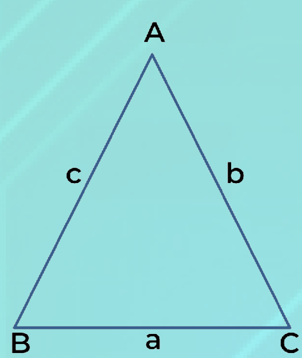
HERON'S FORMULA

KEY POINTS

- In a scalene triangle, if the length of each side is given but its height is not known and it cannot be obtained easily, we take the help of Heron's formula or Hero's formula given by Heron to find the area of such a triangle.

  Heron's formula : If a, b, c denote the lengths of the sides of a triangle ABC. Then,



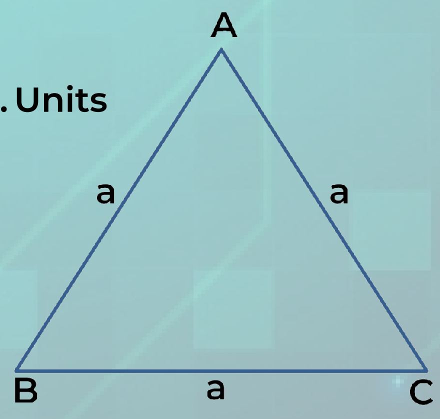
Area of $\Delta ABC = \sqrt{s(s-a)(s-b)(s-c)}$

where s is the semi-perimeter of $\Delta ABC = \dfrac{a+b+c}{2}$

- This formula is applicable to all types of triangles whether it is right-angled or equilateral or isosceles and it is also useful in finding the area of a triangle when it is not possible to find the area of the triangle easily.

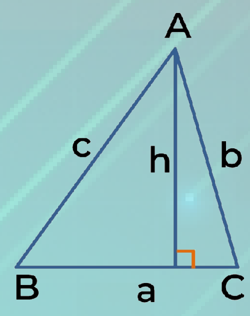
Important formulas

(1) Area of equilateral triangle = $\dfrac{\sqrt{3}}{4} \times \text{Side}^2 = \dfrac{\sqrt{3}}{4} a^2$ Sq. Units

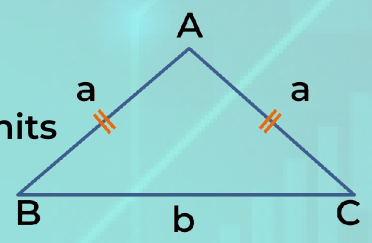


(2) Area of scalene triangle = $\frac{1}{2} \times \text{Base} \times \text{Height}$

$$= \frac{1}{2} \times a \times h \text{ Sq. Units}$$

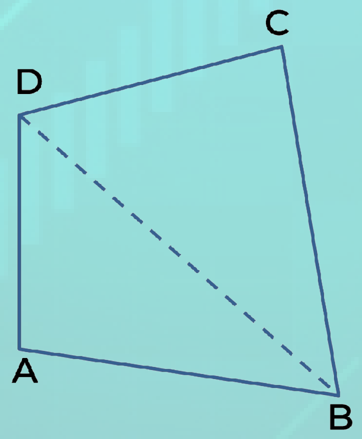


(3) Area of isosceles triangle = $\frac{1}{2} \times \sqrt{\left(a^2 - \frac{b^2}{4}\right)} \times b$ Sq. Units



Applications of Heron's formula in finding area of a quadrilateral

- Heron's formula can be applied to find the area of a quadrilateral by dividing the quadrilateral into two triangular parts.
- Area of each triangle is calculated and the sum of two areas is the area of the quadrilateral.



- Area of quadrilateral ABCD = Area of ΔABD + Area of ΔBCD